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Food and Agriculture Organization of the United Nations

# FAO General Guidelines on Procurement of Works for Field Projects

## Service contracts, Framework Agreements under MS 502 'Procurement of Goods, Works and Services'

***CSLP Guidance Note***

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**References:**

- MS502.19 "Contractual Instruments"
- CSLP intranet: http://intranet.fao.org/csd/procurement/procurement_home

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**Audience**

This guidance note provides guidance to **Buyers** located in Decentralized offices but are also relevant for Requesters, Lead Technical Officers, Technically Competent Officers and Contract Managers involved in projects that have civil works components.

**Purpose**

These guidelines provide colleagues in Field Offices, mainly Buyers, with some practical guidance as regards the **preparation of tenders and contracts** for civil works in line with the Organization terms, conditions and procedures.

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**Introduction**

Construction projects are inherently challenging. Each design is new and each construction site is unique. Success will depend on proper planning, a high level of coordination from parties and individuals that have possibly never worked together before (Architect, Resident Engineer, Contract Manager, Contractor, Sub-contractors, workers and local authorities), the professionalism and integrity of all individuals involved and the environment under which the project is being implemented (security, climate, etc ..).

Construction activities that are not properly planned, resourced or managed are likely to result in cost overruns, poor quality, time delays and programmatic setbacks.

**General Principles**

These Guidelines apply to all civil or construction works undertaken by FAO in Decentralized offices. They apply to all the activities associated with the construction, reconstruction, demolition, repair or renovation of infrastructure and cover all aspects of construction such as civil, mechanical, electrical and associated works.

All works in FAO shall comply, to the extent possible, with the principles of sustainable development by using designs, methodologies, and technical requirements that make efficient use of resources and energy, protect people and ecological systems, maintain and improve the quality of life of the community and benefit their needs.

## Stage 1: Pre-Tender and Tender Stage

### 1. Technical Dossier

Prior to submitting any request to tender for construction works to the Buyer, the Requester must ensure that all the administrative and legal authorizations, to carry out the specific construction works on the

selected piece of land, have been secured by the Requester. These may include inter alia, an environmental impact assessment cleared by the relevant authorities, a valid construction permit, certificate of property of the construction lot, certification by the fire department, authorization to connect to water/electricity/sewage systems, and so forth.

Once the necessary authorizations have been secured, the Requester/Budget Holder shall confirm the availability of funds to cover the costs of the project through the submission of a Purchase Requisition in GRMS together with a complete Technical Dossier to the Buyer so that the procurement process may be initiated. The budget allocated shall be based on an internal estimation of costs taking into account prevailing market prices. If the estimation has been provided by third parties (e.g government, beneficiary), it shall be verified/validated by FAO prior to the solicitation phase.

### 1.1 What should be included in a Technical dossier?

Construction implementation modalities/approaches vary depending on risk allocation, contract value, timeliness and other consideration. For FAO Works engagement, which are often of low or medium value, a traditional approach is used whereby FAO is responsible for preparing the full **Technical dossier,** which provides all the necessary details for potential bidders to quote a fixed amount. With this approach, FAO has control over cost, quality and specifications contrary to other modalities and contracts (e.g Design and Build, etc…) which are more complex, require specialized experience/strong contract management capacity and expose FAO to greater risks of financial overruns.

When the buyer receives the Technical dossier from the Requester/Budget Holder, it should thus contain:

a) **BoQs (Bill of Quantities):** is a document used for tenders in the construction industry in which materials, parts, and labor are itemized by categories of works and serves as a basis for bidders to itemize their costs;

   BoQs should not contain any contingency sum as such amounts are inevitably used by contractors, whether needed for contingencies or not. On the other hand, provisional sums may be considered for low value groups of items when it is easier to include a lump sum in the Bill of Quantities rather than itemizing it (e.g. electrical supplies). In such cases, the original BoQs issued by the Organization will indicate the need to provide a lump sum.

b) **Scope of Works (SoW)**; is a document that includes the purpose of the project, a clear description of the Works, the site location and the key constraints (environment, timeline, etc ..).

c) **Technical specifications*:*** is a document including an explicit set of requirements to be satisfied by a material, design, product, or service. Technical specifications usually refer to standard country specific or internationally recognized building regulations to be applied for the execution of construction and engineering projects;

d) **Drawings*:*** usually consist of architectural plans (e.g. site plan, floor plans, elevation plans and cross sections), structural drawings (e.g. plans and sections, details and notes, foundation plans and notes and structural calculations), mechanical drawings (e.g. plans, sections, calculations and riser diagrams), electrical and lighting drawings (such as floor plans and riser diagrams) and any other graphic information needed for a proper and complete understanding of the project;

e) **Work planning*:*** is a document that involves the choice of technology, the definition of work tasks, the estimation of the required resources and duration for individual tasks, and the identification of any interactions between them. As a minimum, the work planning should specify the commencement date, the order in which the contractor plans to carry out the contract works and a realistic planned completion date.

The technical dossier should preferably be supported by a **topographic study** and if necessary, **geotechnical survey**. A proper justification must be provided in case the topographic study and the geotechnical survey are not carried out.

Similarly, an **Environmental Impact Assessment** might be required depending on the nature of the works and in all cases where Works might have a significant adverse impact on the environment and are subject to a decision of a competent authority (e.g construction of a pesticide store).

### 1.2 Social and Environmental considerations

FAO projects involving Works should seek to avoid both direct and indirect damage to the natural environment and to achieve improvements in the facilities under construction. Sustainability, lifecycle costs, energy use and relevant "Green Issues" should be duly considered at the design/planning stage.

Some of the aspects to be considered are:

. *Environment*: offsetting carbon footprint related to manufacturing and transportation of works material, usage of recyclable or natural materials such as wood with traceable eco labels, preference to local products/materials, design elements - rainwater harvesting, provision for safe waste disposal, use of renewable energy sources, etc…;

. *Social*: employment of local labor, gender inclusive policy in place, etc…;

. *Economic*: lifecycle costs, favoring local entrepreneurs, etc…;

### 1.3 Who should prepare a Technical Dossier?

. FAO staff with civil engineering/architectural background or;

. A consultant with civil engineering/architectural background or;

. A specialized firm: in such case, the company should be selected through a competitive process as per MS 502. If a company is contracted to prepare the technical dossier, this company shall not be invited to bid for the construction itself. The responsibility of the company is to produce a complete technical dossier that, once cleared by the Technically Competent Officer and together with other FAO standard documents, will form part of the Invitation to Bid (ITB). It is not the responsibility of the company to prepare FAO's tender documents and as such, the BoQs produced by the company should not include a section detailing contractual clauses.

Resources available to help prepare your documents are:

a) **Standing Operating Procedures for FAO Field Construction Projects**

http://intranet.fao.org/fileadmin/templates/csa/Infrastructure_Projects/Documents/FAO_Standing_Operating_procedures.pdf

**b) Construction design criteria for FAO buildings**

http://intranet.fao.org/fileadmin/templates/csa/Infrastructure_Projects/Documents/FAO_CSAI_Construction_Design_Criteria.pdf

The Technical Dossier should be cleared by CSLI or by a Technical Officer/Engineer delegated by CSLI to do so.

### 1.4 Risk Management

Risk management is a key element in the procurement of Works. A rigorous risk analysis is required to identify risks related to the delivery of Works especially high value and complex Works and provision made for appropriate risk mitigating measures. In order to mitigate the risks of poor construction that may result from inadequate design, use of poor-quality construction materials, lack of contractor capacity and insufficient quality control during the construction, the following actions should be undertaken:

. All Works design must be undertaken by, or under the technical oversight of, suitably qualified and experienced professionals and cleared by the appropriate local planning agency/authority (if applicable).

. The design/drawings, BoQ, Technical specs and SoW should be cleared by CSLI or by a Technical Officer delegated by CSLI, noting that such clearance will have to be part of the documentation provided to CSLP if the FAO office is requesting an increased Delegation of Authority to carry out the procurement activity.

Procurement in accordance with good practices requires suitable timelines, all the more so for Works where risks are inherent to the nature of the engagement. The sacrificing of diligence for speed due to both external and internal pressures may lead to rushed procurement practices, inevitably resulting in quality, cost and time problems and should be avoided.

## 2. Identification and Role of the Resident Engineer (RE)

The Resident Engineer represents FAO and is, in broad terms, responsible for verifying that Works are carried out in a timely and satisfactory fashion. His/her role and responsibilities are clearly detailed in the standard contract for Works and should be clearly explained to him/her.

The RE should be identified well in advance so as not to delay the start of contract performance. The RE may be a FAO staff with civil engineering/architectural background or a specialized firm, possibly the one that has developed the technical dossier. In the latter case and as indicated above, the firm should be selected from the onset of the project through competitive process as per MS502.

Whilst the Resident Engineer may approve additional Works if so required, authorizing Works that would entail an increase in the value of the contract is under the sole authority of the FAO Procurement Authority and will require a formal amendment to the contract prior to the additional Works being undertaken.

## 3. Invitation to Bid:

### 3.1 Letter of Invitation and Lumpsum Contract

For civil works, FAO typically issues **Invitations to Bid (ITB)** rather than Requests for Proposals as it is expected that the requirements are fully established and described in the Technical Dossier. The ITB is comprised of a **Letter of Invitation with the Technical Dossier** and the **draft Contract**.

Although it might differ from the local practice or the public tenders issued for construction works in the country where FAO is planning to carry out construction works, FAO normally issues "**lump sum**" contracts rather than "re-measurement" contracts for construction works.

Under a lump sum contract, the contract sum is agreed upon before the actual Works begin and **this sum represents the Organization's maximum financial liability**. Payment is then made based on the verification of the completion of work stages rather than on actual measurement of every single BoQ item. In other words, the actual quantities of individual Works might be lower or higher than originally foreseen in the Bill of Quantities, but FAO's financial liability remains the same. This shall be explained to all parties involved (potential bidders in case of pre-bid meeting, Resident Engineer or firm that will carry out the supervision of Works) to avoid any future disagreement at the time of payment.

The detailed BoQs with unit prices and quantities are only produced to ensure all aspects of the construction have been taken into consideration and to easily calculate the financial implication of any possible amendment that may need to be done to the contract if so required and agreed on by the Resident Engineer and Contract Manager.

### 3.2 Handling lots within a single solicitation process

Often there is a need to split the award for similar Works into separable lots or smaller packages either for geographical reasons, to increase supplier diversity, meet programme social objectives and/or mitigate risks of non-performance. The right to split the award between several bidders must be specified in the solicitation document. Due consideration must be given to several factors such as the transaction cost to FAO of placing and managing several contracts as well as local capacity development.

In such cases, potential bidders are likely to bid for more than one lot to increase their chances of selection. The number of lots in a single solicitation should not be more than 3 or 4 to avoid complexity or confusion in the evaluation. **The selection criteria should be clear on the maximum number of lots a bidder can be awarded**, either as an upfront decision to mitigate risk or tied to a bidder's capacity to meet cumulatively lots qualification requirements.

Issuing different bids for similar types of Works at the same time especially if the pool of bidders is small is not recommended. It would be preferable to sequence the issuance of bids.

### 3.3 Pre-bid conference and Site visits

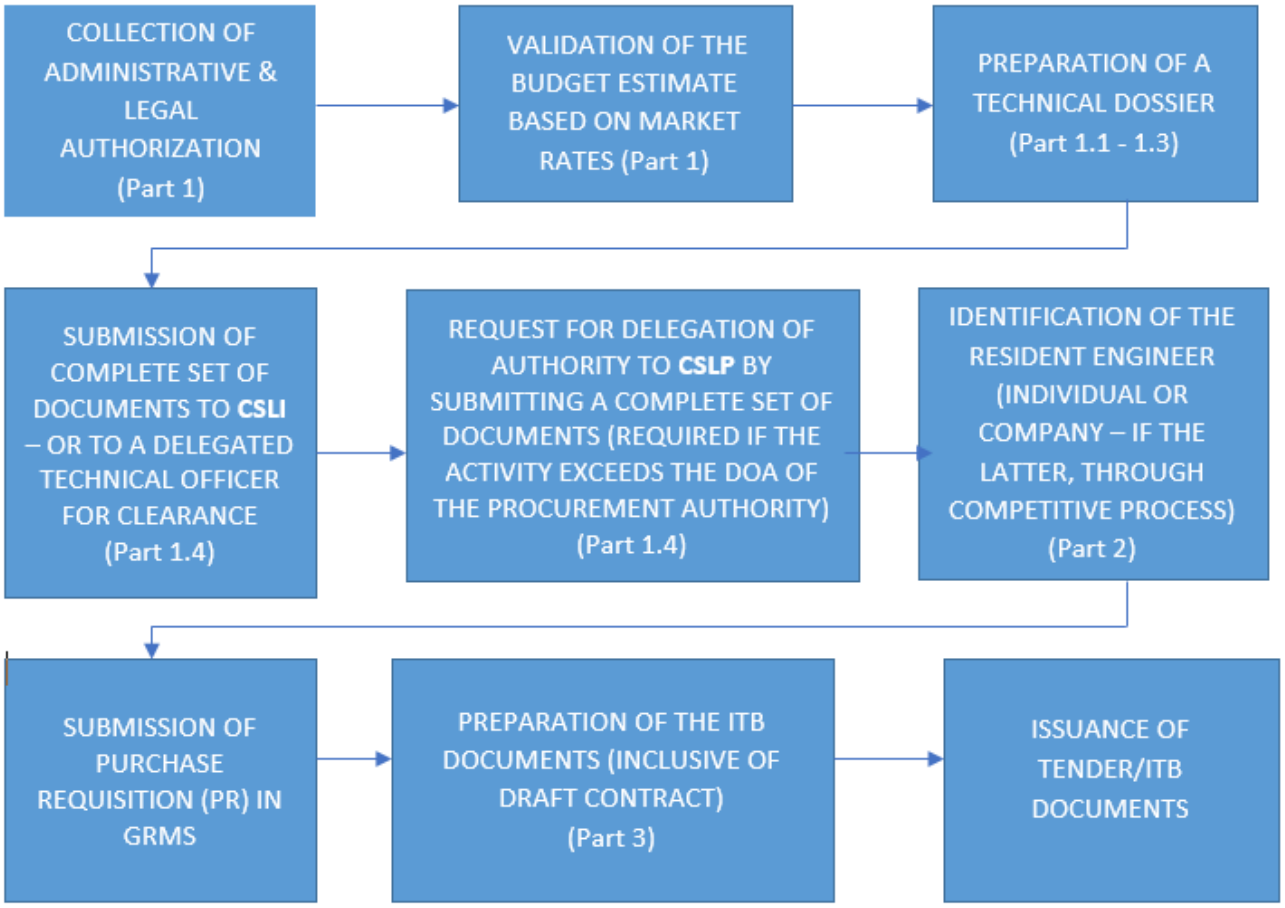
A pre-bid conference may be contemplated to clarify the critical aspects of FAO's Contracts for Works, such as the different roles and milestones established in the contract; nature of the lumpsum contract; guarantees and insurances that must be provided; and processes for payments, retentions and variations.

Site visits are a normal requirement for Works as they give the prospective bidders the opportunity to obtain a clearer and deeper understanding of the site conditions, environment and logistical challenges. Any waiver of site visit should be determined on a case-by-case basis depending on the nature, scope and complexity of the Works as well as potential challenges to access the location.

The ITB should state clearly whether site visit is mandatory with evidence of attendance signed off by a FAO representative and should be at the sole expense of the prospective bidder.

***PROCESS FLOWCHART***

The flowchart below outlines the general process as regards the **preparation of tenders** for civil works.



## 4. Financial Securities

### 4.1 Bank Guarantee (BG) in case of advance payment

The Buyer together with the Contract Manager may decide at the time of the tender to give bidders the possibility to request an advance payment. This should be decided at the tender preparation stage as specific clauses that elaborate the conditions under which an advance payment may be granted need then to be included in the contract that forms part of the tender documents.

When submitting a bid, the Contractor may indeed request an advance payment from the Organization. The advance payment may be up to a **maximum of 20%** of the total contract amount, if this amount is required in order to initiate the Works (e.g. to procure materials). The Organization may only grant the Contractor an advance payment if the Contractor submits a **Bank Guarantee** to the Organization. This Bank Guarantee should be released by a Bank accepted by the Organization and drafted in line with the draft form inserted in the contract.

The cost of the Bank Guarantee is to be borne by the Contractor who should thus consider it when preparing a bid.

### 4.2 Performance Bond (PB)

The Organization may request the Contractor to provide, at the signature of the contract, a Performance Bond valid for the whole duration of the contract **(including the Defects Liability Period**) usually for an amount of **20% of the total amount**, which may be reduced to 10% if used in conjunction with a retention of 10%. The PB is required when dealing with high value contracts or when the Organization wishes to protect itself against unfinished Works or other related risks.

The cost of the Performance Bond is to be borne by the Contractor who should thus consider it when preparing its bid.

### 4.3 Retention

If a financial security cannot easily be obtained (e.g. it is not common in the country, difficulty for the type of bidders invited to have a collateral or to cover the cost of the financial security, etc..), a retention (moneys withheld from contract payment) may be considered as an alternative.

Although the normal industry practice is 5%, a **retention of 10%** may also be considered, noting that increasing the retention percentage may have a negative impact on the Contractor's cash-flow and its ability to complete the Works in a timely and adequate fashion. Smaller payments milestones may be established to reduce this risk.

It is important to note that a retention is only effective for the Organization if the Contractor does undertake the Works. A Performance Bond is a better protection for the Organization in cases where no or

limited Works are eventually completed.

In some cases, the best approach may be to include both a Retention and a Performance Bond. This should be determined on the basis of a simple risk assessment carried out based on the knowledge of the market and environment.

### 5. Defects Liability Period (DLP)

A DLP is a mandatory requirement for work contracts, to enable time after the completion of Works for any defects to become known and corrected by the Contractor. This period which depends on the complexity of the construction and time needed to ascertain possible defects in the Works should be long enough for the Works to pass through one seasonal cycle or a full cycle of use.

The DLP is normally for a period of **twelve (12) months** starting from the date of issuance of the Certificate of (Substantial) Completion of Works. It is thus of paramount importance that the project takes into consideration the Defects Liability Period when designing the project. If project implementation has been delayed and Works (inclusive of DLP) cannot be completed within the project end date, the Buyer should raise the issue from the onset so that a project extension may be requested.

A different DLP period may exceptionally be established depending on the circumstances and nature/complexity of the Works (for example, minor Works such as boundary fencing of gardens), preferably after consulting CSLI. Deviation from the standard twelve-month DLP should be avoided in the interest of FAO.

## Stage 2: Contract Stage

### 1. Contract Manager Role and Responsibilities

Contract management ensures that the selected Contractor performs in accordance with the terms and conditions of the contract. Orderly contract management helps ensure that the Organization receives best value for money.

The Budget Holder either acts as the Contract Manager or is responsible for designating the Contract Manager who will be accountable to the Budget Holder. The Contract Manager is responsible for contract execution and is identified in the contract. The Contract Manager will be supported as necessary and work in collaboration with the Buyer to resolve issues which may arise.

Effective contract management requires clear and open communication between Contract Manager, Resident Engineer, Contractor and as necessary the Buyer. The Contract Manager must remain proactive in identifying potential risks and problems in close collaboration with the Resident Engineer so as to develop a relationship of trust with the Contractor which will lead to satisfactory completion of the project.

Contract Manager responsibilities are listed below and include inter alia:

During contract execution:

- Ensuring financial securities have been provided as per the contract (BG, PB, and so forth);
- Active monitoring of contract implementation and contractor's performance, including costs and schedule controls, conformance with technical specifications and compliance with administrative aspects;
- Identification of breaches in contract performance and discussing remedies with the Contractor;
- Keeping the Buyer and Budget Holder informed about performance issues and escalating issues to the Procurement Authority as necessary;
- Consulting with the Buyer for advice on application of contractual clauses as necessary;
- Requesting amendments of contracts and/or renewals of performance securities as necessary;
- Certifying invoices for payment, including acceptance of Works and handover; and
- Deciding on the application of damages, in consultation with the Buyer, Budget Holder and Procurement Authority as necessary, and notifying the Contractor.

Upon completion of Works:

- Certifying services/Works have been satisfactorily delivered, damages and issues resolved and invoices paid;
- Ensuring documentation regarding acceptance have been duly executed;
- Evaluating Vendor performance using the Supplier Performance Survey, informing the Buyer (in writing) of non-satisfactory performance.

Please consult the Guidance Note on Contract Management for more information.

## 2. Prices

**Unit prices are fixed** for the entire duration of the contract. There can be no adjustment to the financial liability of FAO (contract value) due to fluctuations of market, increase in labour prices or costs of materials, or any currency or interest rate fluctuations.

## 3. Contract Amendment

If, while Works are proceeding, there is a change in the project and FAO deems that additional Works are required (inter alia additional rooms, change in types of materials and so forth), the Contract Manager/Resident Engineer may decide to request an amendment to the contract. This amendment will be based on prices already provided for in the BoQs or on a new quote submitted by the Contractor for item lines which are not part of the BoQs. Due diligence will be applied in reviewing the new quote and comparing them with market rates. Works shall not be undertaken prior to confirmation of availability of funding and issuance of a formal amendment. Proceeding without the latter exposes the Organization to serious financial risks as additional funding might not be available or the Procurement Authority might not agree to the contract amendment.

An amendment is not to be issued however, to cover Works that are already part of the BoQs and/or may

have been overlooked or underestimated by the Contractor in his original bid.

## 4. Financial Securities

### 4.1 Bank Guarantee (BG) in case of advance payment

The Bank Guarantee shall remain in force until FAO has received the counter value in kind of the amount paid in advance.

The advance payment is refunded together with the first invoice or over a few invoices, when the impact is too severe on the Contractor's cash-flow.

### 4.2 Performance Bond (PB)

The Performance Bond must be valid for the whole duration of the contract, i.e. certified completion of Works and expiration of the Defects Liability Period without reservations. It is thus crucial to ensure the validity is extended in cases of contract extensions granted to complete the Works.

The PB is generally reduced by 50% upon Certification of Completion of Works and the remaining 50% extinguished at the end of the Defects Liability Period. A release of a portion of the security must be made in writing.

When the Works have been satisfactorily completed and the Defects Liability Period has expired, the Performance Bond is returned to the bank.

### 4.3 Retention

Half of the amount retained is released upon acceptance of the Certificate of Completion and the remaining 50% at the end of the Defects Liability Period.

## 5. Project Duration

It is of paramount importance that the Contract Manager closely monitors that Works are being carried out according to the agreed timelines. This should be done through regular site visits or through progress or status reports presented by the Resident Engineer. When there are delays, the Contract Manager should initiate a request for extension of the project NTE to accommodate any necessary contract extension, taking into consideration the Defects Liability Period.

This is particularly crucial for projects where funds have to be disbursed rather than committed by the project NTE and going beyond the project NTE may lead to the ineligibility of costs.

## 6. Liquidated Damages (LDs)

Liquidated Damages are a pre-established amount or percentage that the parties agree to as part of the contract. They are usually capped at a percentage of the total contract value when actual damages cannot be objectively determined. The daily or weekly rate can be stated as an actual amount or percentage. The pre-set amount or percentage is an estimate of the value of the loss to the Organization in the event of delay in the contract performance by a Vendor.

Liquidated damages are meant to be a compensation for loss, not a penalty.

To be able to apply LDs on partial payments, the contract must have a detailed construction program which enables verification of the delay. In case of delay, the Contract Manager / Resident Engineer will have notified the Contractor in writing and given the opportunity to the company to remedy before applying any LDs, noting that such a decision may have an impact on the Contractor's cash flow and may thus lead to further delay.

The Contract Manager will have to exercise best judgment and his/her knowledge of the market and environment to determine what the impact of applying Liquidated Damages will be and whether and how they should be applied. A record should then be kept of such decisions.

Please consult Guidance Note on Liquidated Damages for more information.